

CS 4700/6700 Programming Project

Fall 2026

Due day: December 4th, Friday

Simulation of Extendible Hashing.

1. The hash value of the hash-key attribute of each record is created as a randomly generated integer value.
2. Each data bucket (aka logical block) can store 40 records (actually their hash values) and is dynamically allocated. Each directory entry stores a pointer to a data bucket.
3. Generate up to 10,000 hash values (to simulate up to 10,000 records), and for every 100 hash values, compute and display followings:
 - $\text{Utilization_of_the_directory} = \frac{\text{total_number_of_data_buckets}}{\text{total_number_of_entries_in_the_directory}}$
 - $\text{Utilization_of_the_data_buckets} = \frac{\text{total_number_of_records_stored_in_the_data_buckets}}{\text{maximum_number_of_records_that_can_be_stored_in_all_the_data_buckets}}$
4. You can use any programming language: C, C++, Java, Python, etc.
5. Your final report should include the description of your design, source code of your program, print out of execution results (i.e., above utilizations), and discussion.
 - You can plot the execution results (i.e., above utilizations) if you want.
6. Grading: Your final report will be graded based on {design 5%, documentation 5%, correctness 15%, discussion 5%}
7. References:
 - Section 16.8.3 of the text book.
 - R. Fagin, J. Nievergelt, N. Pippenger, and H. R. Strong, "Extendible Hashing: A fast access method for dynamic files," ACM Trans. on Database Systems, vol. 4, no. 3, 1979, pp. 315–344.